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EE4760 – Smart Buildings and Green Technology

1. Specific Energy Consumption (SEC) of the building (kWh/m²/year)

$$SEC = \frac{\text{Annual Total Building Energy (kWh)}}{\text{Total Floor Area (m}^2\text{)}} = \frac{2,037.3 \times 10^3 \text{ kWh}}{40 \times 40 \times 6 \text{ m}^2} = 212.218 \frac{\text{kWh}}{\text{m}^2} / \text{year}$$

Annual Energy Consumption by Enduse

	Electricity kWh (x000)	Natural Gas Btu	Steam Btu	Chilled Water Btu
Space Cool	536.8	-	-	-
Heat Reject.	29.8	-	-	-
Refrigeration	-	-	-	-
Space Heat	-	-	-	-
HP Supp.	-	-	-	-
Hot Water	-	-	-	-
Vent. Fans	92.3	-	-	-
Pumps & Aux.	262.3	-	-	-
Ext. Usage	33.5	-	-	-
Misc. Equip.	752.9	-	-	-
Task Lights	-	-	-	-
Area Lights	329.8	-	-	-
Total	2,037.3	-	-	-

2. Cooling power consumption

$$\text{Cooling power consumption} = \frac{\text{Annual Cooling Energy (kWh)}}{\text{Total Floor Area (m}^2\text{)}} = \frac{1,175,006 \text{ kWh}}{40 \times 40 \times 6 \text{ m}^2} = 122.396 \frac{\text{kWh}}{\text{m}^2} / \text{year}$$

Office Building - Colombo

DOE-2.2-47h2 5/13/2026 10:48:29 BDL RUN 5

REPORT- SS-D Building HVAC Load Summary

WEATHER FILE- Los Angeles CA TMY2

- - - - - C O O L I N G - - - - -						- - - - - H E A T I N G - - - - -						- - - E L E C - - -		
MONTH	COOLING ENERGY (MBTU)	TIME OF MAX DY HR	DRY- BULB TEMP	WET- BULB TEMP	MAXIMUM COOLING LOAD (KBTU/HR)	HEATING ENERGY (MBTU)	TIME OF MAX DY HR	DRY- BULB TEMP	WET- BULB TEMP	MAXIMUM HEATING LOAD (KBTU/HR)	ELEC- TRICAL ENERGY (KWH)	MAXIMUM ELEC LOAD (KW)		
JAN	49.55088	26 14	82.F	55.F	1787.487	-0.047	3 8	53.F	42.F	-9.447	94346.	348.911		
FEB	56.09909	12 14	85.F	60.F	1933.199	-0.042	21 12	61.F	51.F	-8.237	88837.	348.118		
MAR	75.47161	31 16	75.F	53.F	1900.508	-0.010	28 8	53.F	52.F	-3.868	102063.	347.241		
APR	105.91557	13 14	77.F	61.F	2101.329	-0.006	25 8	56.F	53.F	-2.896	102421.	353.516		
MAY	233.59656	29 16	74.F	65.F	2969.483	0.000	31 1	61.F	57.F	0.000	94830.	345.754		
JUN	316.91837	30 15	75.F	64.F	2776.791	0.000	30 1	64.F	61.F	0.000	102173.	348.395		
JUL	549.98285	20 16	74.F	67.F	3504.209	0.000	31 1	65.F	60.F	0.000	102613.	349.755		
AUG	612.42670	17 15	75.F	68.F	4119.633	0.000	31 1	62.F	59.F	0.000	99422.	349.232		
SEP	481.32880	8 9	72.F	66.F	3500.041	0.000	30 1	64.F	60.F	0.000	98103.	349.885		
OCT	328.37399	22 15	73.F	68.F	3066.977	0.000	31 1	59.F	55.F	0.000	99010.	350.194		
NOV	131.68567	10 13	73.F	62.F	2012.969	-0.005	28 8	53.F	44.F	-5.089	89454.	348.945		
DEC	92.99192	18 14	76.F	56.F	1756.078	-0.060	12 8	46.F	36.F	-28.268	101730.	345.885		
TOTAL	3034.342					-0.170					1175006.			
MAX					4119.633					-28.268		353.516		
MAXIMUM DAILY INTEGRATED COOLING LOAD (DES DAY)						0.000 (KBTU)								
MAXIMUM DAILY INTEGRATED COOLING LOAD (WTH FILE)						37170.744 (KBTU)								